



SAPIENZA
UNIVERSITÀ DI ROMA

Modellazione

MIP

Volume V

Problems to model

Forty problems given as they really arrive — a text, some data, a question — with no model already written: twenty with explicit numerical data and twenty in symbolic form. The work of recognising the structure is left to be done. The solutions are reserved for instructors.

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Online version and downloads:

fabiofurini.github.io/modellazione-mip/
github.com/fabiofurini/modellazione-mip

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Twenty numerical problems to model

The problems of this chapter have their data written out, like the fifteen numerical models of volume II, but they come without the model. One reads the statement, recognises the decisions, writes the MILP and solves it.

The steps are always the same:

1. list the **decisions** and give each one a variable with its domain;
2. turn every sentence of the statement into a **constraint**;
3. check the **links** between variables with the truth table;
4. build by hand a **feasible solution** and, where possible, a **dual solution**, to get the two bounds;
5. solve the model and **compare**.

The solutions are reserved for instructors and are not published.

N1 — Six projects and a budget

N1 — the projects to fund

A department chooses which projects to fund out of six. The costs are 40, 25, 30, 15, 50 and 20 thousand euros; the expected benefits are 9, 5, 7, 3, 11 and 4. The budget is 100. Projects 2 and 5 use the same laboratory and cannot both be funded; project 6 only makes sense if project 1 is funded too. The maximum benefit is wanted.

Modelling question 1.1 — model and optimum

Write the model and say how many constraints it has as a function of the data. What is the optimum of the instance?

N2 — Four couriers, five deliveries

N2 — the morning deliveries

Five deliveries must be given to four couriers. The time courier i takes for delivery j , in minutes, is

	D1	D2	D3	D4	D5
courier 1	20	35	25	40	30
courier 2	25	20	30	35	45
courier 3	30	25	20	30	25
courier 4	40	30	35	20	35

Each delivery goes to one courier only; each courier works at most 60 minutes. The minimum total time is wanted.

Modelling question 1.2 — model and optimum

Write the model. What changes if the time of the busiest courier is to be minimised instead?

N3 — Antennas over five districts

N3 — the minimum coverage

A city has five districts and four possible antenna sites. Site 1 covers districts $\{1, 2\}$, site 2 covers $\{2, 3, 4\}$, site 3 covers $\{1, 4, 5\}$ and site 4 covers $\{3, 5\}$. Every antenna costs the same. All districts must be covered with the smallest number of antennas.

Modelling question 1.3 — model, optimum and bound

Write the model and solve it. Then write the dual of the LP relaxation and build a feasible solution of it by hand: which bound does it give?

N4 — Seven parcels in ten-kilo boxes

N4 — how many boxes are needed

Seven parcels weigh 6, 5, 4, 4, 3, 2 and 2 kilos. Each box carries at most 10 kilos and a parcel is not split. The smallest number of boxes is wanted.

Modelling question 1.4 — model, optimum and combinatorial bound

Write the model. Give a lower bound without solving the MILP, and say whether it is attained.

N5 — Two products, three departments

N5 — the integer product mix

A workshop makes two articles. One piece of the first takes 2 hours in department A, 1 in B and 3 in C; one piece of the second takes 1, 3 and 2. The departments have 20, 24 and 30 hours available. The unit margins are 7 and 6 euros, and pieces are sold whole only. The maximum margin is wanted.

Modelling question 1.5 — model, optimum and price of integrality

Write the model. How much is the linear relaxation worth, and what does integrality cost on this instance?

N6 — The shifts of the week

N6 — the counter clerks

A counter is open seven days. The clerks needed are, from Monday to Sunday, 4, 3, 4, 5, 6, 5 and 3. Every clerk hired works five consecutive days and then rests two; the cycle may start on any day of the week. The smallest number of clerks is wanted.

Modelling question 1.6 — model and optimum

Write the model. How many constraints does it have, and how many variables?

N7 — Three suppliers with a set-up charge

N7 — how much to buy from whom

100 units of a component are needed. Three suppliers sell it at 9, 7 and 8 euros a unit, but a supplier that receives an order also charges a fixed 50, 120 and 90 euros. The first supplier cannot exceed 40 units, the second 70, the third 60. The minimum spend is wanted.

Modelling question 1.7 — model, big-M and optimum

Write the model. What is the smallest M that can be used for each supplier, and why is it valid?

N8 — Five jobs on one machine

N8 — the worst lateness

Five jobs run one after the other on a single machine. The durations are 4, 2, 6, 3 and 5 hours; the due dates are 8, 5, 14, 10 and 18 hours from the start. The machine starts at time 0 and never stops. The worst tardiness — the maximum between zero and completion minus due date — is to be minimised.

Modelling question 1.8 — model and optimum

Write the model with sequencing variables. What changes if the sum of the tardinesses is wanted instead?

N9 — Four warehouses and six customers**N9 — where to open**

Six customers ask for 12, 8, 15, 10, 6 and 9 pallets. Four warehouses may be opened, with capacities 30, 25, 20 and 35 pallets and opening costs 100, 90, 80 and 120. The transport cost per pallet from warehouse l to customer c is

	c1	c2	c3	c4	c5	c6
warehouse 1	2	4	5	7	6	3
warehouse 2	3	2	4	6	5	4
warehouse 3	5	3	2	4	3	6
warehouse 4	6	5	3	2	2	5

The minimum total cost is wanted.

Modelling question 1.9 — model and optimum

Write the model. What is the link between opening and transport, and in which direction does the constraint impose it?

N10 — Six students in three groups**N10 — the balanced groups**

Six students have average marks 28, 24, 30, 22, 26 and 25. They must be split into three groups of two. The group with the highest average and the one with the lowest should be as close as possible.

Modelling question 1.10 — model and optimum

Write the model with the difference between maximum and minimum. How much is the LP relaxation worth, and why?

N11 — Five films in two screens**N11 — the evening programme**

Five films last 90, 120, 100, 140 and 110 minutes. Two screens are free for 240 minutes each. Every film is shown at most once, and in one screen only. The expected takings are 500, 700, 450, 800 and 600 euros. Films 2 and 4 have the same audience and must not be programmed in the same screen. The maximum takings are wanted.

Modelling question 1.11 — model and optimum

Write the model. Which link does the constraint on films 2 and 4 impose, and what happens if it is written as $x_{2s} + x_{4s} \leq 1$ for every screen s ?

N12 — Diet with four foods**N12 — the weekly ration**

Four foods cost 2, 3, 1 and 4 euros a kilo. One kilo of each provides

	f1	f2	f3	f4
protein (g)	20	35	10	40
iron (mg)	3	2	5	4

At least 120 grams of protein and 25 milligrams of iron are needed. A food, if it enters the ration, enters for at least half a kilo. The minimum spend is wanted.

Modelling question 1.12 — model and minimum lot

Write the model. What happens to the solution if the half-kilo constraint is dropped?

N13 — Purchases in brackets**N13 — the quantity discount**

A firm buys up to 200 units of a material. The price is 10 euros a unit for the first 50, 8 for those between 51 and 120, 7 beyond 120. The discount applies only to the units in the bracket, not to all of them. At least 90 units are needed, and the store holds at most 180. The minimum spend is wanted.

Modelling question 1.13 — piecewise model

Write the model with one variable per bracket and the adjacency constraints that prevent using a bracket before the previous one is full.

N14 — Six junctions to watch**N14 — the cameras at the junctions**

A road network joins six junctions with the stretches $\{1,2\}$, $\{1,3\}$, $\{2,3\}$, $\{2,4\}$, $\{3,5\}$, $\{4,5\}$, $\{4,6\}$ and $\{5,6\}$. A camera at a junction watches every stretch meeting there. All stretches must be watched with the smallest number of cameras.

Modelling question 1.14 — model, optimum and bound

Write the model. Give a lower bound by looking at pairwise disjoint stretches.

N15 — Three months of production

N15 — set-ups and stocks

A line makes one article for three months. The demands are 100, 140 and 80 pieces. Making a piece costs 5 euros; starting production in a month costs 300 euros whatever the quantity; holding a piece at the end of a month costs 1 euro. The line makes at most 150 pieces a month. Stock starts and ends empty. The minimum cost is wanted.

Modelling question 1.15 — model, big-M and optimum

Write the model. What is the smallest valid M for each month?

N16 — Four containers

N16 — the compatible load

Four containers carry 20, 15, 25 and 18 tonnes. Six loads weigh 10, 8, 12, 6, 9 and 14 tonnes. Loads 1 and 3 are incompatible and do not travel in the same container; load 6 needs a refrigerated container, and only 2 and 4 are. Every load must be shipped. Using a container costs 100 euros. The minimum cost is wanted.

Modelling question 1.16 — model and optimum

Write the model. How many families of constraints are needed, and what does each one say?

N17 — Six activities with precedences

N17 — the building site

Six activities last 3, 2, 4, 1, 5 and 2 days. Activity 3 starts only after 1 and 2; 5 after 3; 6 after 4 and 5. Two crews work in parallel, and an activity occupies a crew for its whole duration. Finishing as early as possible is wanted.

Modelling question 1.17 — model and makespan

Write the model with start times and the disjunctions between activities on the same crew. Which lower bound can be read from the data without solving?

N18 — Five emergency crews

N18 — the skills

Five crews must be formed from a pool of ten technicians. Every technician has one of three skills: technicians 1–4 have the first, 5–7 the second, 8–10 the third. Every crew has two technicians and must have two different skills. Every technician is in one crew only. Technicians 3 and 9 do not work together. Whether a formation exists is to be decided.

Modelling question 1.18 — model and feasibility

Write the model. If it is infeasible, say which constraint makes it so and prove it by counting.

N19 — Three periods with backlog**N19 — delivering late has a price**

The demands of three periods are 50, 70 and 40 units. Production costs 4 euros a unit and does not exceed 60 units per period. Holding a unit at the end of a period costs 1 euro; delivering a unit late costs 3 euros per period of delay. Everything must be delivered by the end of the third period. The minimum cost is wanted.

Modelling question 1.19 — model and balance

Write the balance with stocks and backlog, and say why the two variables cannot both be positive at the optimum.

N20 — Eight pictures on two walls**N20 — the hanging**

Eight pictures are 60, 45, 80, 50, 70, 40, 55 and 65 centimetres wide. Two walls are 240 centimetres long each. Every picture must be hung, and on one wall only. The free space on the two walls should be as equal as possible.

Modelling question 1.20 — model and optimum

Write the model minimising the difference between the free space of the two walls. How much is the LP relaxation worth, and why is it useless?

Twenty symbolic problems to model

Here the data are not numbers but symbols, declared with their type and their unit, as in the twenty-three problems of volume III. Every problem brings two or three links between variables into play: the model has to be written in general, with the quantifiers, and for every family of constraints one has to say how many constraints there are as a function of the data. As everywhere in the course, the data are written with Latin letters: Greek letters are reserved for the dual variables.

For each of them one is asked for:

1. the **decisions** and the variables, with domain and count;
2. the complete **model**, with quantified objective and constraints;
3. for every family of constraints, **which sentence** of the statement it translates and **how many constraints** it is;
4. the **links** between variables appearing in it, with the direction imposed by the constraint and the one imposed by optimality;
5. a **feasible solution** built by hand and a **dual solution** of the relaxation, with the two bounds.

The solutions are reserved for instructors and are not published.

S1 — Projects with a budget, exclusions and a portfolio bonus

S1 — the funding plan

A body must choose which projects to fund among $n \in \mathbb{Z}_{\geq 1}$ candidates. For every project $j \in \{1, 2, \dots, n\}$, the value $c_j \in \mathbb{Q}_{>0}$ is the cost, in euros, and $p_j \in \mathbb{Q}_{>0}$ is the expected return, in euros. The budget is $b \in \mathbb{Q}_{>0}$ euros. The set E contains the pairs $\{i, j\}$ of projects that use the same laboratory and cannot both be funded; the set D contains the ordered pairs (i, j) such that funding i forces funding j as well. A subset $S \subseteq \{1, 2, \dots, n\}$ gathers the strategic projects: if at least $\ell \in \mathbb{Z}_{\geq 1}$ of them are funded, the body receives an extra contribution of $\bar{r} \in \mathbb{Q}_{>0}$ euros, once only. The body wants to decide which projects to fund, at maximum total return.

S2 — Tasks and resources to activate

S2 — the assignment with a minimum load

A department must give $n \in \mathbb{Z}_{\geq 1}$ tasks to $m \in \mathbb{Z}_{\geq 1}$ resources. For every resource $i \in \{1, 2, \dots, m\}$ and every task $j \in \{1, 2, \dots, n\}$, the value $c_{ij} \in \mathbb{Q}_{>0}$ is the cost, in euros, of giving task j to resource i , and $t_{ij} \in \mathbb{Q}_{>0}$ is the time that task takes on that resource, in hours. For every resource i , the value $b_i \in \mathbb{Q}_{>0}$ is the time available, in hours; $f_i \in \mathbb{Q}_{>0}$ is the fixed activation cost, in euros, to be paid if the resource receives at least one task; $\ell_i \in \mathbb{Q}_{>0}$ is the minimum load, in hours, that an activated resource must reach, with $\ell_i \leq b_i$. Every task goes to exactly one resource. The department wants to decide which resources to activate and how to distribute the tasks, at minimum total cost.

S3 — Posts with a service radius and a backup

S3 — the coverage and the second post

An administration must cover $m \in \mathbb{Z}_{\geq 1}$ users by opening posts in some of $n \in \mathbb{Z}_{\geq 1}$ candidate sites. For every site $j \in \{1, 2, \dots, n\}$, the value $f_j \in \mathbb{Q}_{>0}$ is the opening cost, in euros, and $u_j \in \mathbb{Z}_{\geq 1}$ is the largest number of users the post can take in charge. For every site j and every user $i \in \{1, 2, \dots, m\}$, the value $d_{ij} \in \mathbb{Q}_{\geq 0}$ is the distance, in kilometres; the site may serve the user only if $d_{ij} \leq r$, where $r \in \mathbb{Q}_{>0}$ is the service radius. Every user must be taken in charge by exactly one open post. The users of a critical subset K must in addition have a second open post within the radius, even if it does not take them in charge. The administration wants to decide where to open and who takes whom in charge, at minimum total cost.

S4 — Containers of several types with incompatible goods

S4 — what travels together

A warehouse must store $n \in \mathbb{Z}_{\geq 1}$ objects in containers of $k \in \mathbb{Z}_{\geq 1}$ types. For every object $j \in \{1, 2, \dots, n\}$, the value $w_j \in \mathbb{Q}_{>0}$ is the weight, in kilos, and $g_j \in \{1, 2, \dots, q\}$ is the goods class, among $q \in \mathbb{Z}_{\geq 1}$ classes. For every container type $h \in \{1, 2, \dots, k\}$, the value $c_h \in \mathbb{Q}_{>0}$ is the capacity, in kilos, $e_h \in \mathbb{Q}_{>0}$ is the cost of one container of that type, in euros, and $N_h \in \mathbb{Z}_{\geq 1}$ is the number available. The set F contains the pairs $\{a, b\}$ of classes that cannot share a container. An object is not split. The warehouse wants to decide how many containers of each type to use and how to fill them, at minimum total cost.

S5 — Lots with set-up, capacity and a limited store

S5 — production over n periods

A firm plans the production of one article over $n \in \mathbb{Z}_{\geq 1}$ periods. For every period $t \in \{1, 2, \dots, n\}$, the value $d_t \in \mathbb{Q}_{\geq 0}$ is the demand, in pieces; $p_t \in \mathbb{Q}_{>0}$ is the cost of making one piece, in euros; $q_t \in \mathbb{Q}_{>0}$ is the fixed set-up cost of the line, in euros, to be paid if anything is made in that period; $c_t \in \mathbb{Q}_{>0}$ is the largest quantity that can be made in the period, in

pieces; $v_t \in \mathbb{Q}_{>0}$ is the smallest quantity made if anything is made, in pieces, with $v_t \leq c_t$. For every period $t \in \{1, 2, \dots, n-1\}$, the value $h_t \in \mathbb{Q}_{>0}$ is the cost of holding one piece at the end of the period, in euros; the store holds at most $M \in \mathbb{Q}_{>0}$ pieces, in every period. The values $r_0 \in \mathbb{Q}_{\geq 0}$ and $r_n \in \mathbb{Q}_{\geq 0}$ are the initial stock and the one required at the end, in pieces. The firm wants to decide how much to make in every period, at minimum total cost.

S6 — p sites, capacity and the worst distance

S6 — the radius to minimise

A firm must serve $m \in \mathbb{Z}_{\geq 1}$ customers by opening exactly $p \in \mathbb{Z}_{\geq 1}$ sites among $n \in \mathbb{Z}_{\geq 1}$ candidates, with $p \leq n$. For every customer $c \in \{1, 2, \dots, m\}$, the value $d_c \in \mathbb{Q}_{>0}$ is the demand, in pallets. For every site $l \in \{1, 2, \dots, n\}$, the value $u_l \in \mathbb{Q}_{>0}$ is the capacity, in pallets; for every site l and customer c , the value $t_{lc} \in \mathbb{Q}_{\geq 0}$ is the distance, in kilometres. Every customer is served entirely by one open site, and an open site does not exceed its capacity. A subset A of sites is already owned by the firm; among those not owned, at most $\bar{p} \in \mathbb{Z}_{\geq 1}$ may be opened. The firm wants to decide where to open and who serves whom, minimising the largest distance between a customer and the site serving it.

S7 — Qualified machines and release dates

S7 — the completion of the last job

A workshop must run $n \in \mathbb{Z}_{\geq 1}$ jobs on $k \in \mathbb{Z}_{\geq 1}$ machines. For every job $j \in \{1, 2, \dots, n\}$, the value $t_j \in \mathbb{Q}_{>0}$ is the processing time, in hours, the same on every machine; $r_j \in \mathbb{Q}_{\geq 0}$ is the release date, that is the instant before which the job cannot start; $Q_j \subseteq \{1, 2, \dots, k\}$ is the set of machines qualified to run it. For every machine $i \in \{1, 2, \dots, k\}$, the value $a_i \in \mathbb{Q}_{\geq 0}$ is the instant from which the machine is free. Every machine runs one job at a time, and a job, once started, is not interrupted. The workshop wants to decide on which machine and in which order to run the jobs, minimising the instant at which the last one finishes.

S8 — Shelves with classes and a maximum weight

S8 — the heaviest shelf

A depot must distribute $n \in \mathbb{Z}_{\geq 1}$ objects over $m \in \mathbb{Z}_{\geq 1}$ shelves. The objects belong to $q \in \mathbb{Z}_{\geq 1}$ goods classes: for every object $j \in \{1, 2, \dots, n\}$, the value $g_j \in \{1, 2, \dots, q\}$ is the class and $w_j \in \mathbb{Q}_{>0}$ is the weight, in kilos. Every shelf may hold objects of at most $s \in \mathbb{Z}_{\geq 1}$ different classes, with $s \leq q$, and carries at most $c \in \mathbb{Q}_{>0}$ kilos. The objects of a subset P are fragile and must go on shelves that are all different. The depot wants to decide the distribution, minimising the weight of the heaviest shelf.

S9 — Cutting with several bar formats

S9 — the pieces and the waste

A workshop must obtain pieces of $k \in \mathbb{Z}_{\geq 1}$ different lengths from bars of $h \in \mathbb{Z}_{\geq 1}$ formats. For every format $f \in \{1, 2, \dots, h\}$, the value $L_f \in \mathbb{Q}_{>0}$ is the length of the bar, in metres, $g_f \in \mathbb{Q}_{>0}$ is the cost of one bar, in euros, and $N_f \in \mathbb{Z}_{\geq 1}$ is the number available. For every length $i \in \{1, 2, \dots, k\}$, the value $\ell_i \in \mathbb{Q}_{>0}$ is the length of the piece, in metres, and $b_i \in \mathbb{Z}_{\geq 1}$ is the number of pieces required. The total waste cannot exceed a fraction $e \in \mathbb{Q}_{>0}$, with $e < 1$, of the total length of the bars used. The workshop wants to decide how many bars of each format to use and how to cut them, at minimum total cost.

S10 — Network with a fixed charge per arc and a limit on arcs

S10 — the flow and the arcs to activate

A logistics operator must send $Q \in \mathbb{Q}_{>0}$ units of goods from an origin node s to a destination node u over a directed network of nodes V and arcs $A \subseteq V \times V$; the nodes other than s and u are transit nodes. For every arc $(i, j) \in A$, the value $c_{ij} \in \mathbb{Q}_{>0}$ is the cost of sending one unit, in euros; $f_{ij} \in \mathbb{Q}_{>0}$ is the fixed activation cost, in euros; $k_{ij} \in \mathbb{Q}_{>0}$ is the capacity, in units. The activated arcs cannot be more than $\bar{a} \in \mathbb{Z}_{\geq 1}$, and out of every transit node at most one activated arc may leave. The operator wants to decide which arcs to activate and how much to send on them, at minimum total cost.

S11 — Cyclic shifts with rest and skills

S11 — the staff of the service

A service runs over $n \in \mathbb{Z}_{\geq 1}$ days, on a weekly cycle. The staff is divided into $q \in \mathbb{Z}_{\geq 1}$ skills. For every day $t \in \{1, 2, \dots, n\}$ and every skill $c \in \{1, 2, \dots, q\}$, the value $b_{tc} \in \mathbb{Z}_{\geq 0}$ is the number of people of that skill who must be on duty that day. For every skill c , the value $w_c \in \mathbb{Q}_{>0}$ is the weekly cost of one person, in euros, and $m_c \in \mathbb{Z}_{\geq 0}$ is the number of people of that skill already hired. Whoever works is on duty $\ell \in \mathbb{Z}_{\geq 1}$ consecutive days and then rests $g \in \mathbb{Z}_{\geq 1}$, and the cycle may start on any day. The service wants to decide how many people to hire per skill and with which starting day, at minimum total cost.

S12 — Portfolio in lots with sectors

S12 — the funds to buy

An investor has a capital of $K \in \mathbb{Q}_{>0}$ euros and chooses among $n \in \mathbb{Z}_{\geq 1}$ funds, divided into $k \in \mathbb{Z}_{\geq 1}$ sectors. For every fund $j \in \{1, 2, \dots, n\}$, the value $s_j \in \{1, 2, \dots, k\}$ is the sector, $q_j \in \mathbb{Q}_{>0}$ is the lot size, in euros — the fund is bought in whole lots only — and $r_j \in \mathbb{Q}_{>0}$ is the expected return per euro invested. In no sector may more than a fraction $a \in \mathbb{Q}_{>0}$, with $a \leq 1$, of the capital be invested. At least $p \in \mathbb{Z}_{\geq 1}$ different funds must be chosen, and a chosen fund receives at least $v_j \in \mathbb{Z}_{\geq 1}$ lots. The investor wants to decide how many lots of each fund to buy, at maximum total return.

S13 — A ration over several days

S13 — the canteen and the stocks

A canteen prepares the ration of $n \in \mathbb{Z}_{\geq 1}$ days using $s \in \mathbb{Z}_{\geq 1}$ foods and respecting $r \in \mathbb{Z}_{\geq 1}$ nutritional constraints. For every food $i \in \{1, 2, \dots, s\}$, the value $w_i \in \mathbb{Q}_{>0}$ is the cost of one kilo, in euros; $c_i \in \mathbb{Q}_{>0}$ is the smallest quantity, in kilos, used on a day where the food appears; $u_i \in \mathbb{Q}_{>0}$ is the largest daily quantity, with $c_i \leq u_i$; and $h_i \in \mathbb{Q}_{>0}$ is the cost of holding one kilo in the pantry from one day to the next, in euros. For every food i and nutrient $j \in \{1, 2, \dots, r\}$, the value $g_{ij} \in \mathbb{Q}_{\geq 0}$ is the amount of nutrient j in one kilo of food i ; for every nutrient j , every day at least $a_j \in \mathbb{Q}_{>0}$ and at most $b_j \in \mathbb{Q}_{>0}$ are needed. Every day at least $t \in \mathbb{Z}_{\geq 1}$ different foods must appear. The canteen wants to decide how much to buy and how much to use each day, at minimum total cost.

S14 — Sequences with sequence-dependent set-ups and due dates

S14 — the order of the jobs

On a single machine $n \in \mathbb{Z}_{\geq 1}$ jobs must run. For every job $j \in \{1, 2, \dots, n\}$, the value $t_j \in \mathbb{Q}_{>0}$ is the processing time, in hours, and $d_j \in \mathbb{Q}_{>0}$ is the due date, that is the instant by which the job should be completed; delivering late costs $w_j \in \mathbb{Q}_{>0}$ euros per hour of delay. For every pair of distinct jobs i and j , the value $s_{ij} \in \mathbb{Q}_{\geq 0}$ is the set-up time, in hours, needed to go from job i to job j . The machine starts at time 0, runs one job at a time and does not stop during a job. The workshop wants to decide the order of the jobs, minimising the total cost of the delays.

S15 — Crews over several periods

S15 — covering the shifts

A service company must cover $m \in \mathbb{Z}_{\geq 1}$ zones for $n \in \mathbb{Z}_{\geq 1}$ periods, with $k \in \mathbb{Z}_{\geq 1}$ crews. For every crew $j \in \{1, 2, \dots, k\}$, the value $c_j \in \mathbb{Q}_{>0}$ is the cost of employing it for one period, in euros, and $S_j \subseteq \{1, 2, \dots, m\}$ is the set of zones it can cover. In every period every zone must be covered by at least one employed crew, and a crew employed in a period covers all and only the zones of S_j . A crew cannot be employed for more than $\ell \in \mathbb{Z}_{\geq 1}$ periods out of n , and if it is employed it is employed for at least $v \in \mathbb{Z}_{\geq 1}$ periods, with $v \leq \ell$. The company wants to decide which crews to employ and in which periods, at minimum total cost.

S16 — Students and supervisors with balanced loads

S16 — assigning the theses

A degree programme must assign $n \in \mathbb{Z}_{\geq 1}$ students to $m \in \mathbb{Z}_{\geq 1}$ supervisors. For every student $i \in \{1, 2, \dots, n\}$ and every supervisor $j \in \{1, 2, \dots, m\}$, the value $v_{ij} \in \mathbb{Z}_{\geq 0}$ is the preference score the student gives that supervisor. For every supervisor j , the value $u_j \in \mathbb{Z}_{\geq 1}$ is the largest number of students they can follow. Every student is assigned to exactly one supervisor, and nobody may receive a supervisor to whom they gave a score below $\bar{v} \in \mathbb{Z}_{\geq 0}$.

The programme wants to decide the assignment maximising the sum of the preferences, with the constraint that the difference between the number of students of the busiest supervisor and that of the least busy among those with at least one student does not exceed $g \in \mathbb{Z}_{\geq 0}$.

S17 — Maintenance with windows and lost production

S17 — when to stop the machines

A plant has $n \in \mathbb{Z}_{\geq 1}$ machines and plans the maintenance over $T \in \mathbb{Z}_{\geq 1}$ periods. For every machine $j \in \{1, 2, \dots, n\}$, the values $a_j \in \{1, 2, \dots, T\}$ and $b_j \in \{1, 2, \dots, T\}$, with $a_j \leq b_j$, delimit the window in which the maintenance must be done, once only; $p_j \in \mathbb{Q}_{>0}$ is the hourly output of the machine, in pieces; $\ell_j \in \mathbb{Z}_{\geq 1}$ is the duration of the maintenance, in consecutive periods. For every machine j and period t , the value $c_{jt} \in \mathbb{Q}_{>0}$ is the cost of the maintenance crew, in euros, if the maintenance starts in that period. In no period may more than $k \in \mathbb{Z}_{\geq 1}$ machines be stopped, and the total lost output cannot exceed $M \in \mathbb{Q}_{>0}$ pieces. The plant wants to decide when to stop each machine, at minimum total cost.

S18 — Balanced partition of a graph

S18 — two groups, few links

An organisation must split into two teams the $n \in \mathbb{Z}_{\geq 1}$ nodes of an undirected graph with edge set E . For every node $i \in \{1, 2, \dots, n\}$, the value $w_i \in \mathbb{Q}_{>0}$ is the workload it carries; for every edge $\{i, j\} \in E$, the value $u_{ij} \in \mathbb{Q}_{>0}$ is the intensity of the exchange between the two nodes. The total load of each team must lie between $L \in \mathbb{Q}_{>0}$ and $U \in \mathbb{Q}_{>0}$, with $L \leq U$. Some pairs of nodes, gathered in the set P , must end up in the same team; others, gathered in F , in different teams. The organisation wants to decide the split, minimising the total intensity of the exchanges between the two teams.

S19 — Loading with heterogeneous vehicles

S19 — what goes on what

A carrier must ship $n \in \mathbb{Z}_{\geq 1}$ loads with $m \in \mathbb{Z}_{\geq 1}$ vehicles. For every load $j \in \{1, 2, \dots, n\}$, the value $w_j \in \mathbb{Q}_{>0}$ is the weight, in kilos, $o_j \in \mathbb{Q}_{>0}$ is the volume, in cubic metres, and $V_j \subseteq \{1, 2, \dots, m\}$ is the set of vehicles it may go on. For every vehicle $i \in \{1, 2, \dots, m\}$, the value $c_i \in \mathbb{Q}_{>0}$ is the payload, in kilos, $s_i \in \mathbb{Q}_{>0}$ is the volume available, in cubic metres, and $f_i \in \mathbb{Q}_{>0}$ is the cost of using it, in euros. The set E contains the pairs $\{a, b\}$ of loads that cannot travel on the same vehicle. Every load goes on exactly one vehicle. The carrier wants to decide which vehicles to use and how to split the loads, at minimum total cost.

S20 — Capacity to install with bracketed costs

S20 — the technologies and the brackets

An operator must install a total capacity of at least $Q \in \mathbb{Q}_{>0}$ megawatts choosing among $n \in \mathbb{Z}_{\geq 1}$ technologies. For every technology $j \in \{1, 2, \dots, n\}$, the value $k_j \in \mathbb{Q}_{>0}$ is the capacity of one installed unit, in megawatts, and the units are installed in whole numbers; the unit cost falls in brackets: for every bracket $h \in \{1, 2, \dots, m_j\}$, with $m_j \in \mathbb{Z}_{\geq 1}$, the value $u_{jh} \in \mathbb{Z}_{\geq 1}$ is the number of units in the bracket and $c_{jh} \in \mathbb{Q}_{>0}$ is the cost of one unit in that bracket, with $c_{j1} > c_{j2} > \dots > c_{jm_j}$; a bracket is used only once the previous ones are full. At most $p \in \mathbb{Z}_{\geq 1}$ different technologies may be used, and a technology that is used is installed for at least $v_j \in \mathbb{Z}_{\geq 1}$ units. The operator wants to decide how much of each technology to install, at minimum total cost.